

Customer Data Foundation for Retention Analysis

Field-level customer data mapping for regulatory completeness, profile reliability, and product-level retention analysis

Context KoinWorks fintech reporting	Scope Borrowers + multi-source profile data	Tools BigQuery, SQL, Tableau, Confluence	Output SQL view -> Airflow data mart table
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Business problem

Customer profile data was fragmented across multiple operational and enrichment sources, creating gaps in mandatory borrower fields and reducing confidence in regulatory, credit-risk, and downstream reporting outputs.

Credit-bureau and regulatory reporting required mandatory borrower profile fields, but the customer source of truth was fragmented across borrower/lender warehouse tables, loan and transaction data, OCR ID-card output, JSON records in the data lake, and spreadsheet-based re-KYC enrichment. Missing identity and profile fields - especially ID number, DOB, gender, phone number, and province - could block outstanding-borrower submission and reduce confidence in downstream reporting.

What I built

A BigQuery-based customer profile layer that applied field-level survivorship rules across borrower, loan, transaction, OCR/KYC, JSON, and re-KYC sources, then supported borrower segmentation and churn analysis by product, demographic, and activity status.

Business value

Improved customer profile reliability for recurring reporting and gave stakeholders a cleaner way to separate acquisition-heavy product growth from stronger retention-quality segments.

My approach: field-level survivorship logic

I treated the customer data mart as a field-level source-of-truth problem rather than assuming one table was correct for every attribute. The main customer/borrower entity became the base layer, then each important field was evaluated against source ownership, completeness, freshness, validation quality, and conflict risk.

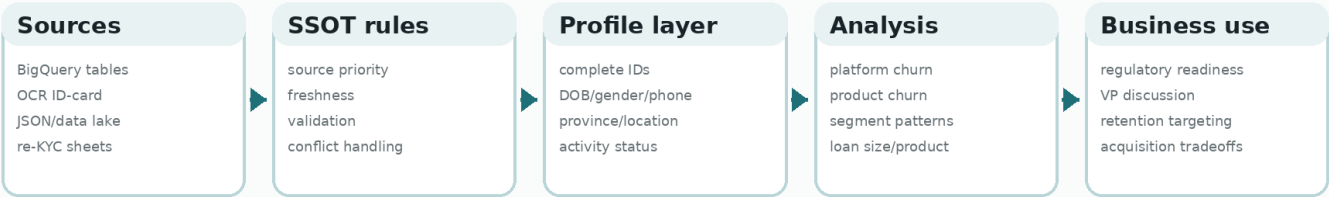
Field type	Survivorship rule	Examples
Stable identity fields	Prioritize verified/KYC/OCR/re-KYC sources and validate before filling or overriding; conflicts require review.	ID/KTP number, DOB, gender
Mutable contact/location fields	Use freshness and verified updates more heavily, while checking completeness and valid formats.	phone number, province/address
Product / relationship fields	Use the operational source that owns loan, product, transaction, and activity status logic.	active loan, outstanding balance, renewal, last activity
Derived fields	Derive missing attributes from trusted reference logic and document the transformation.	postal code -> province; ID number -> DOB/gender/location

Validation checks and reporting output

Data quality checks Null completeness, duplicate IDs, invalid KTP format, invalid phone format, conflicting KTP/name/DOB, inconsistent city/province naming checked against postal-code reference data	Final reporting output Cleaned customer profile SQL view first, then ingested by data engineering through Airflow into a data mart table analysts could use as a shared customer profile layer
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Customer data foundation to churn segmentation workflow

Source completeness first; product and demographic retention insight second.



Churn and customer segmentation layer

Using the cleaned profile layer to separate acquisition growth from retention quality

Churn definitions

Platform churn Customer had no active loan and no transaction/activity for the last 3 months.	Product churn Product relationship was no longer active, outstanding, or renewed for a specific product.
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Segmentation logic

The analysis focused on borrowers and separated overall churn from product-level churn so product growth could be evaluated more carefully. Segments included age, gender, province, income group, employee versus business-owner profile, loan size, and product type. Product groups included short-term consumer lending and business-owner lending, alongside other loan/investment products.

Analytical findings and stakeholder use

Acquisition vs retention Short-term consumer-loan promotions could support profit or acquisition, but also attracted more one-time-user churn.	Retention opportunity Business-owner loan customers showed stronger retention/renewal potential and became a better candidate for retention focus.
Product-level view Platform churn and product churn told different stories; one metric could hide whether users left the platform or only cycled out of one product.	Stakeholder discussion Findings were presented at a high level to a VP and used to support marketing discussions around product retention and acquisition strategy.

Recommendation framing

The recommendation was to avoid treating all product growth equally: separate acquisition-driven promotions from retention-quality signals, monitor one-time-user churn for short-term lending promotions, and prioritize retention strategy for customer/product segments with stronger renewal potential. Follow-up observations were directionally consistent with the segmentation: business-owner lending appeared to show stronger renewal quality, while short-term lending promotions looked more acquisition-driven and required churn monitoring. These are presented as observed directional outcomes, not causal proof.

Project value and limitations

This project demonstrates work below the dashboard layer: mapping messy source systems, defining field-level source-of-truth rules, improving mandatory-field completeness for regulatory reporting, and turning a more reliable profile layer into product and demographic retention analysis.

- Directional churn analysis only; causal impact would require campaign exposure, controlled cohorts, and consistent retention windows.
- Public materials should anonymize product labels and avoid exposing internal table names, customer values, or submission details.
- Future improvements: source-trace fields, completeness-score monitoring, and cohort-level profit/retention tracking by product segment.